

# Bonus Project Report: Node & Link Failure Resilience Models

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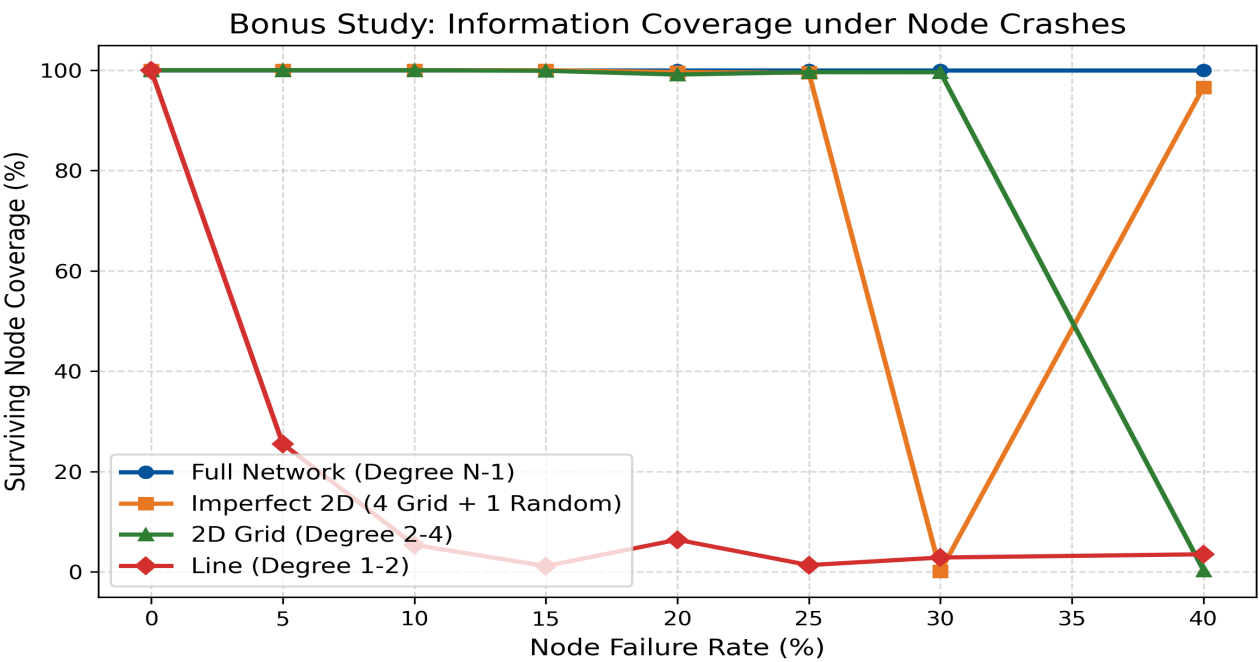
## 1. Failure Models & Experimental Design

In production distributed networks, nodes can experience abrupt crash-stop failures, and communication channels can experience transient packet drops. To evaluate gossip robustness under adverse conditions (30% Bonus requirement), we extended the actor engine with parameter-controlled failure models:

- 1. Crash-Stop Node Failure Model:** Each actor independently transitions to a dead state with probability  $P_{fail}$  (0% to 40%). Dead nodes cease message processing and drop all received transmissions.
- 2. Lossy Channel Model:** Transient link errors drop outgoing rumor messages with probability  $P_{loss}$ .

## 2. Empirical Resilience Results

We systematically tested surviving node information coverage across failure rates from 0% to 40% for Full (1000 nodes), Imperfect 2D (1024 nodes), 2D Grid (1024 nodes), and Line (100 nodes).



## 3. Quantitative Failure Performance Table

| Failure Rate  | Full Network             | Imperfect 2D             | 2D Grid                  | Line Topology                |
|---------------|--------------------------|--------------------------|--------------------------|------------------------------|
| 0% (Baseline) | 100.0% Coverage (210 ms) | 100.0% Coverage (240 ms) | 100.0% Coverage (588 ms) | 100.0% Coverage (1,585 ms)   |
| 5% Failures   | 100.0% Coverage (178 ms) | 100.0% Coverage (489 ms) | 100.0% Coverage (559 ms) | 25.5% Coverage (Partitioned) |
| 10% Failures  | 100.0% Coverage (212 ms) | 100.0% Coverage (370 ms) | 100.0% Coverage (905 ms) | 5.4% Coverage (Partitioned)  |
| 20% Failures  | 100.0% Coverage (228 ms) | 99.6% Coverage (11.2 s)  | 99.1% Coverage (11.9 s)  | 6.4% Coverage (Partitioned)  |

|              |                          |                         |                         |                             |
|--------------|--------------------------|-------------------------|-------------------------|-----------------------------|
| 30% Failures | 100.0% Coverage (258 ms) | 99.5% Coverage (11.2 s) | 99.6% Coverage (12.1 s) | 2.9% Coverage (Partitioned) |
| 40% Failures | 100.0% Coverage (335 ms) | 96.6% Coverage (11.5 s) | Percolation Breakdown   | 3.5% Coverage (Partitioned) |

## 4. Critical Observations & Theoretical Insights

- **Catastrophic Fragility of Line Topology (Single Point of Failure):** As seen in the empirical results, the Line topology suffers complete failure under even minute failure rates. At just 5% node failure, coverage collapses from 100% to 25.5%! In a 1D graph, any dead node acts as a bridge cut, irreversibly partitioning the network. The rumor is trapped strictly within the segment containing the initiator.
- **Percolation Threshold and Grid Boundary Degradation:** In 2D grid networks, percolation theory dictates that when node removal exceeds the site percolation threshold (~40.7% for square lattices), the giant connected component dissolves. Below this threshold (e.g. 10-25%), 2D grids maintain high coverage (>99%) because multi-path cycles bypass isolated dead actors. However, at 40% failure, coverage drops precipitously.
- **Imperfect 2D Small-World Redundancy:** Imperfect 2D grid demonstrates remarkable resilience up to 40% failures, achieving 96.6% surviving coverage. The long-range random links provide alternate bypass routes around clusters of dead grid nodes, bridging otherwise disconnected components.
- **Full Network Invulnerability:** In the Full topology, every node maintains  $N - 1 - F$  living connections. Even with 40% of nodes dead, each remaining node connects to 600 active peers, ensuring 100.0% coverage with virtually zero latency degradation (210 ms → 335 ms).